

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

19th May 2023

**Subject CP1 – Actuarial Practice
(Paper B)**

Time allowed: 3 Hours 15 Minutes (14.45 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.***
- 2. *Attempt all questions beginning your answer to each question on a separate sheet.***
- 3. *Mark allocations are shown in brackets.***
- 4. *Please check if you have received complete Question Paper and no page is missing. If so kindly get new set of Question Paper from the Invigilator.***

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1) A local white goods manufacturing company wants to purchase business interruption cover for its main manufacturing plant situated on the outskirts of a major industrial city.

i) Highlight the benefits to the manufacturer of purchasing the business interruption cover. (4)

The manufacturer has approached one of the reputed general insurers to provide the business interruption cover for the plant. The general insurer wants to conduct risk assessment of the manufacturing plant before issuing the business interruption cover.

ii) Outline all the information with explanation the insurer may require for understanding the risks and estimating the potential loss before providing the cover to the manufacturing plant. (13)

The white goods manufacturer provides life cover to the employees of the manufacturing plant through Yearly Renewable Group Term Life plan purchased through ABC Life Insurance Co. Ltd. The renewal of the plan is due next month and hence it has asked the existing life insurer to provide premium quotation for the group life term plan. The quote received by the manufacturer has seen a significant increase in the premium rate over the last year.

iii) What are the likely reasons for increase in the premium rates obtained by the manufacturer? (7)

The CFO of the manufacturer has proposed to the CEO that the company self-insure rather than go for the Group Term Life plan and reassess the situation next year. According to the CFO this will help them save costs as there will not be any need to pay high premiums to purchase the cover from life insurer.

iv) Discuss the merits and demerits of self-insuring the cover by the manufacturer. (6)

The CRO of ABC Life Insurance Co. Ltd. has been asked to put in place a liquidity management framework and monitor & report the liquidity position of the insurer to the risk committee on monthly basis.

v) Briefly discuss the likely stressed scenarios which can impact the current and the likely future liquidity position of the insurer? (6)

ABC Life Insurance Co. Ltd. has a reasonable amount of equity investments in the non-participating fund and the shareholder fund. The CRO is concerned about the extremely high equity market valuations, as any market correction will impact the balance sheet and likely threaten the solvency of the life insurer.

vi) Describe different strategies that could be used to protect the value of the equity investment and the risk as result of implementing those strategies. (7)

CFO of ABC Life Insurance Co. Ltd is determining the amount of dividend which it can pay for the recently concluded financial year.

vii) Describe the factors that the CFO would need to consider in determining the amount of dividend that it can pay to the shareholder? (7)

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Q. 2) A Fast-Moving Consumer Goods (FMCG) major corporate entity with country-wide operations is planning to introduce a Defined Benefit (DB) pension scheme for its employees. It is proposed that the benefit payable will be 50% of the average salary drawn by the employees in their final year of service before retirement. The pension will be payable monthly. Upon the death of the pensioner, 50% of the last drawn monthly pension is proposed to be payable to the surviving spouse. It is also proposed to pay a lumpsum benefit to the beneficiaries upon the death of the pensioner.

- i) Briefly discuss the reasons why the benefits received by the employees after retirement will be less valuable / inadequate and the reasons why the employer may not be able to provide the promised benefits. (8)

The company has approached the Actuarial consultancy firm you work for, to model the benefits and calculate provisions of the proposed scheme.

- ii) Explain in detail the key steps involved in developing a model for a DB pension scheme. (8)

- iii) List the cashflows that are modelled in a DB Pension scheme. (4)

- iv) List down the assumptions used in the model and explain how the assumptions are set. (6)

The Chief Risk Officer (CRO) is concerned about the overall impact of the proposed pension scheme on the financial position of the company.

- v) What do you think are the major concerns of the CRO related to the proposed pension scheme and how the same could be mitigated? (10)

As on date, pension schemes in the country are not funded. However, the Government is in the process of introducing a new regulation that may make funding mandatory for all the pension schemes.

- vi) Discuss the merits and demerits of funded pension schemes. (6)

The corporate entity is a public limited company listed on the stock exchange and hence required to disclose salient features of the pension scheme along with the annual accounts.

- vii) Briefly discuss why disclosure is required and the other requirements that will be reported for the benefit of major stakeholders of the company. (8)

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